

Dynamic Fire Evacuation Router with Real-Time Hazard Mapping

Idea Presentation

- Problem Statement ID – 2
- Problem Statement Title – Dynamic Fire Evacuation Router with Real-Time Hazard Mapping
- Theme – IoT and Real-Time Embedded Systems for Life Safety
- PS Category – Software
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Proposed solution

Idea title: Dynamic Fire Evacuation Router with Real-Time Hazard Mapping

- Decentralized, MQTT-based system senses, fuses, and reroutes live
- Non-linear sensor fusion feeds one centralized Dijkstra routing pass
- Broadcasts safest exit direction to every zone within 300 ms
- Replaces static exit signage with live-recalculated LED guidance
- Falls back to shelter-in-place when no safe path exists
- Validated under real load testing, not just a single demo

Technical approach

- Python: Flask, NetworkX, paho-mqtt for core routing logic
- Mosquitto MQTT broker; Node-RED for monitoring dashboard
- HTML, CSS, JavaScript control panel; Web Audio API alarm
- Hardware-ready: ESP32 with DHT22, MQ-2, and IR sensors

Figure 1. Sensor-to-dashboard system pipeline



Feasibility and viability

Fully working software prototype validated end-to-end across 36 zones.

Issue	Root cause	Resolution
Redundant recomputation	Each of 36 nodes independently re-ran Dijkstra per update	One centralized coordinator computes and broadcasts routes
MQTT publish volume growth	Every node republished LED state on every broadcast	Publish only on real state change, plus keepalive

Result: typical recompute latency of 20–130 ms, within the 300 ms budget.

Artifacts

- Source code: full repository publicly available at github.com/kishaunjith-S/fire-evac-router

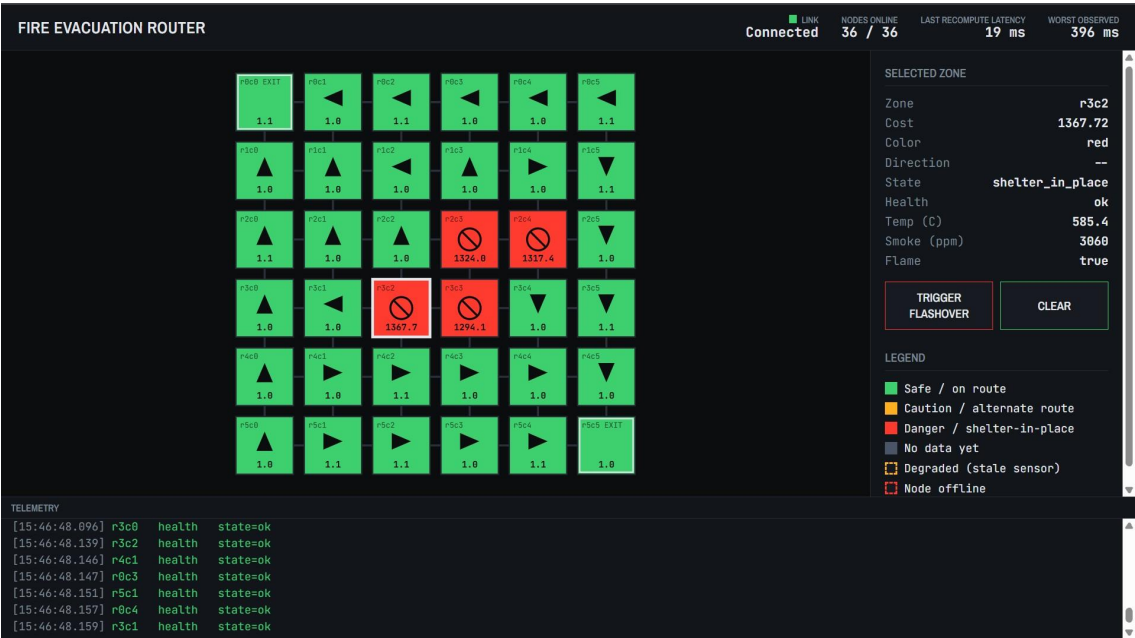


Figure 2. Control-panel simulator

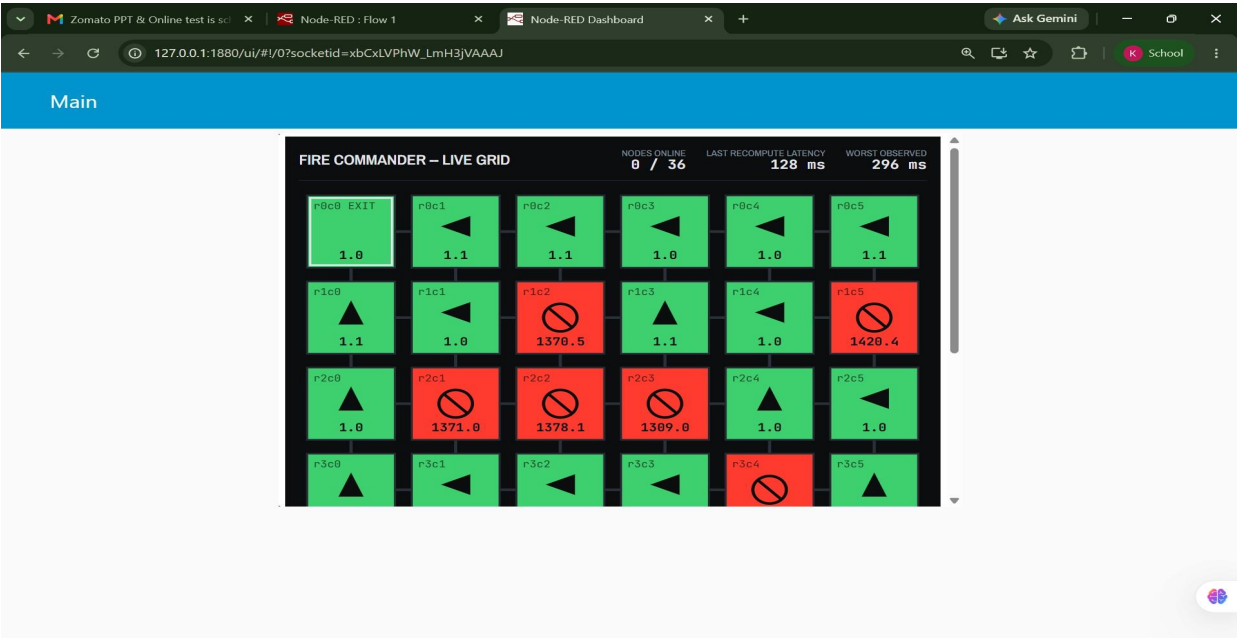


Figure 3. Fire Commander dashboard (Node-RED)

Research and references

- Repository: github.com/kishaunjith-S/fire-evac-router
- Debugging case study: [docs/latency_debugging_notes.md](#)
- Libraries: NetworkX, Eclipse Mosquitto, paho-mqtt, Node-RED
- Calibration sources: NIST fire data, Kaggle smoke datasets